



Indian Institute of Technology Gandhinagar

Semester-II, Academic Year: 2025-26

ORDINARY DIFFERENTIAL EQUATIONS (MA 104) : FINAL EXAM

21/40

Date: 23/02/2026

Max. Mark 40

Duration : 9:00 am - 11 am

Weightage: 40 %

- (i) Please write clearly and *provide proper justification* for your statements.
- (ii) Books, notebooks, calculators, mobile phones, and other electronic gadgets are not allowed during the examination.
- (iii) Answer one question from Question 6 or 7 and one from Question 8 or 9. If more than one question is attempted, only the first answer will be evaluated.

- (1) Use Picard's method of successive approximations to solve the initial value problem

$$y e^{x/2} = \int x e^{x/2}$$

$$y' + xy = x, \quad y(0) = 0.$$

$$y_{n+1} = \text{circled } y_0 + \int_0^{x_n} f(t, y_n) dt$$

- (2) (a) Given that $y_1(x) = \frac{\cos x}{x}$ is a solution of the differential equation $xy'' + 2y' + xy = 0$, find a second linearly independent solution by reducing the order of the equation.

- (b) Consider the initial value problem $y' = \frac{e^{y^2}-1}{1-x^2y^2}$ with $y(-2) = 1$ on the rectangle $D = (-5/2, 3/2) \times (3/4, 5/4) \subseteq \mathbb{R}^2$. Find h such that a unique solution exist on the interval $|x+2| < h$ and justify your choice. [3+3]

- (3) (a) Use the method of undetermined coefficients to solve $y'' + 10y' + 25y = 14e^{-5x} (q + c_2x + r x^2) e^{-5x}$
- (b) Use the method of variation of parameters to solve $y''' + y' = \sec t$ with $-\frac{\pi}{2} < t < \frac{\pi}{2}$.

- (4) (a) Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of $x^2y'' + 5xy' + 4y = x^2 \sin x$. [2+4]

Given $W(y_1(2), y_2(2)) = \frac{1}{32}$, find the Wronskian $W(y_1(x), y_2(x))$.

- (b) Consider the ODE $y''' + p_2y'' + p_1y' + p_0y = 0$. Let $\lambda_1, \lambda_2, \lambda_3$ be the roots of the characteristic equation $\lambda^3 + p_2\lambda^2 + p_1\lambda + p_0 = 0$. Show that $e^{\lambda_1x}, e^{\lambda_2x}, e^{\lambda_3x}$ form a basis for the solution space if and only if $\lambda_1, \lambda_2, \lambda_3$ are distinct. [3+3]

- (5) (a) Write the Rodrigues's formula for the Legendre polynomial $P_n(x)$ of degree n . Show that for $m < n$ as well as $m - n = 2k + 1$, where k is a non-negative integer,

$$\int_{-1}^1 x^m P_n(x) dx = 0.$$

You may assume that $\frac{d^k}{dx^k}(x^2 - 1)^n \Big|_{x=\pm 1} = 0$ for all $k < n$.

- (b) Using the generating function $G(x, t) := \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n$ of the Legendre polynomials $\{P_n(x)\}_{n \geq 0}$ prove that $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$. [1+3+2]

- (6) (a) Find a general solution of $x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0$ in terms of the Bessel functions J_ν and $J_{-\nu}$, using the substitution $\lambda x = z$.

- (b) Use the properties $[x^\nu J_\nu(x)]' = x^\nu J_{\nu-1}(x)$ and $[x^{-\nu} J_\nu(x)]' = -x^{-\nu} J_{\nu+1}(x)$ to show

(i) $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$

(ii) $J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x)$

$\frac{d}{dx} (x^{n+1} J_{n+1}(x)) / x^n$

Use the above properties to prove $xJ'_{n+1}(x) + (n+1)J_{n+1}(x) = xJ_n(x)$. [3+3]

- (7) Let $b(x)$ and $c(x)$ be analytic functions at $x = 0$. Consider $x^2 y'' + b(x)xy' + c(x)y = 0$.

Suppose the roots $r_1 \geq r_2$ of the indicial equation satisfy $r_1 - r_2 \notin \mathbb{Z}$.

- (a) Find a fundamental set of solutions $\{y_1(x), y_2(x)\}$ with explicit recurrence relation for the coefficients of each one.

- (b) Show that $x^2 y'' - \frac{x}{2} y' + \frac{1+x}{2} y = 0$ satisfies the hypotheses and compute the first three nonzero terms of $y_1(x)$ and $y_2(x)$ for this differential equation. [3+3]

- (8) State and prove the existence theorem for the Laplace transform. Verify that the following functions

(i) $f(t) = [t]$, where $[t] = n$ for $n \leq t < n+1$ with $n \in \mathbb{Z}$.

(ii) $f(t) = t^2 e^{-t}$.

$|f(t)| \leq M e^{at}$

satisfy the hypotheses of the theorem. [1+3+1+1]

- (9) Use Laplace transforms to solve the initial value problem

$$y'' + 4y = \begin{cases} 8 \sin t, & 0 < t < \pi, \\ 0, & t > \pi. \end{cases} \quad \text{with } y(0) = 4, \quad y'(0) = 4,$$

[6]

$\sum_{i=0}^{m-1} \left(\frac{1}{2} a_{m-i} + (i+\sigma) b_i a_{m-i} \right) \rightarrow \sigma_1 \rightarrow y_1$

$m = \frac{\dots}{(m+\sigma)^2 + (b_0 - 1)(m+\sigma) + c_0} \rightarrow \sigma_2 \rightarrow y_2$